[21] In the former edition and Appendix, I have given some facts on the transportal of erratic boulders and icebergs in the Atlantic Ocean. This subject has lately been treated excellently by Mr. Hayes, in the Boston Journal (vol. iv. p. 426). The author does not appear aware of a case published by me (Geographical Journal, vol. ix. p. 528) of a gigantic boulder embedded in an iceberg in the Antarctic Ocean, almost certainly one hundred miles distant from any land, and perhaps much more distant. In the Appendix I have discussed at length the probability (at that time hardly thought of) of icebergs, when stranded, grooving and polishing rocks, like glaciers. This is now a very commonly received opinion; and I cannot still avoid the suspicion that it is applicable even to such cases as that of the Jura. Dr. Richardson has assured me that the icebergs off North America push before them pebbles and sand, and leave the submarine rocky flats quite bare; it is hardly possible to doubt that such ledges must be polished and scored in the direction of the set of the prevailing currents. Since writing that Appendix, I have seen in North Wales (London Phil. Mag., vol. xxi. p. 180) the adjoining action of glaciers and floating icebergs.

CHAPTER XII

CENTRAL CHILE

Valparaiso—Excursion to the Foot of the Andes—Structure of the Land—Ascend the Bell of Quillota—Shattered Masses of Greenstone—Immense Valleys—Mines—State of Miners—Santiago—Hot-baths of Cauquenes—Gold-mines—Grinding-mills—Perforated Stones—Habits of the Puma—El Turco and Tapacolo—Humming-birds.

JULY 23rd.—The Beagle anchored late at night in the bay of Valparaiso, the chief seaport of Chile. When morning came, everything appeared delightful. After Tierra del Fuego, the climate felt quite delicious—the atmosphere so dry, and the heavens so clear and blue with the sun shining